Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/21/2006
PI Name:	Conley, Catherine A Ph.D.		
Project Title:	A New Model for Understanding Spaceflight Induced Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-0234
Organization Name:	NASA - AMES Research Center		
PI Address 1:	SLR		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:	NOTE (June 2016): Dr. Conley is currently at NASA Headquarters		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	07/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jacobson, Lewis (University of Pittsburgh)		
Grant/Contract No.:	ARC Intramural PI; NNA04CK22A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this project is to develop the nematode C. elegans as a small animal model for studying the molecular basis of spaceflight induced muscle atrophy. C. elegans is an established model both for studying the molecular basis of muscle atrophy on Earth and for studying spaceflight effects on physiology. The principal challenge addressed during this project period is the development, characterization and evaluation of a chemically defined liquid culture system that allows automation of C. elegans experimentation and analysis in flight. We have broken this work into three Aims for the current project period: 1) Adapt the C. elegans muscle protein degradation model to liquid culturing; 2) Validate liquid culturing of C. elegans in the spaceflight environment; 3) Determine biological responses of C. elegans in liquid culture to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) We have developed a system for the automated culture of the common laboratory animal C. elegans. The system can be used as an enviornmental biomonitor, in pharmaceutical discovery, and in compound toxicity testing. 2) Based upon our work with worm muscles, which lack satellite cells, we believe that altered muscle development may be a contributing factor to human muscle pathologies (in the form of altered development of satellite cells which fuse with developed muscle). If true, this finding opens new avenues of investigation toward muscle pathology treatments. 3) We have identified new genetic triggers of and signaling pathways which regulate protein degradation in worm muscle. These pathways may represent targets for new treatments of human muscle pathologies.
Task Progress:	Summary: Our timetable called for completing Aim 2 in grant years 1-2 and Aims 1 and 3 in grant years 1-3. Aim 2 is now complete and work on Aims 1 and 3 continues. Two peer-reviewed articles are currently in press, two invited talks were given, and two public outreach talks were given. Two manuscripts are currently being written and a workshop is planned for later this year. Progress on the Aims: Aim 1: Adapt the C. elegans muscle protein degradation model to liquid culturing. (a) Do mutant, transgenic strains of C. elegans grow in CeMM? We have demonstrated that the mutant strains grow. In the process, we have determined that gene expression is altered in response to altered diet. A manuscript outlining the gene expression changes and the consequent biological changes is currently being prepared and will be submitted for publication in a peer reviewed journal. A collaboration with Dr. Judith L. Fridovich-Keil of Emory University School of Medicine was established this year to examine the detailed function of one of the genes whose expression is altered in CeMM-grown worms. Severe mutations of this gene in humans reportedly result in mental retardation and infertility even in treated individuals. (b) Do the same factors trigger and block muscle protein degradation in strains cultured in CeMM as on NGM? We have demonstrated this is true by pharmacological and/or mutant experiments during long-term cultivation in CeMM. In the process, we have found that muscle protein degradation is triggered in response to altered diet (i.e., acute shift from NGM to CeMM). We have utilized our existing mutants to determine that the signals that trigger protein degradation in response to acute dietary shift are at least partially independent of the known signaling pathways by which degradation is induced in response to starvation, or excess Fibroblast Growth Factor signaling or deficiency of Insulin/Insulin like Growth Factor. We have also determined that muscle protein degradation in response to altered diet can be blocked pharmacologically with agents that stimulate muscle contraction (nicotinic receptor agonist) or with an inhibitor of the protein-degrading proteasome. We plan to continue to investigate the molecular mechanism by which muscle protein degradation is controlled by diet and activity.
Bibliography Type:	Description: (Last Updated: 04/15/2008)
Articles in Peer-reviewed Journals	Higashitani A, Higashibata A, Sasagawa Y, Sugimoto T, Miyazawa Y, Szewcyk NJ, Viso M, Gasset G, Eche B, Fukui K, Shimazu T, Fujimoto N, Kuriyama K, Ishioka N. "Checkpoint and physiological apoptosis in germ cells proceeds normally in spaceflown Caenorhabditis elegans." Apoptosis. in press 2005. , Jun-2005
Articles in Peer-reviewed Journals	Szewczyk NJ, Jacobson LA. "Signal-transduction networks and the regulation of muscle protein degradation." Int J Biochem Cell Biol. in press 2005. , Jun-2005